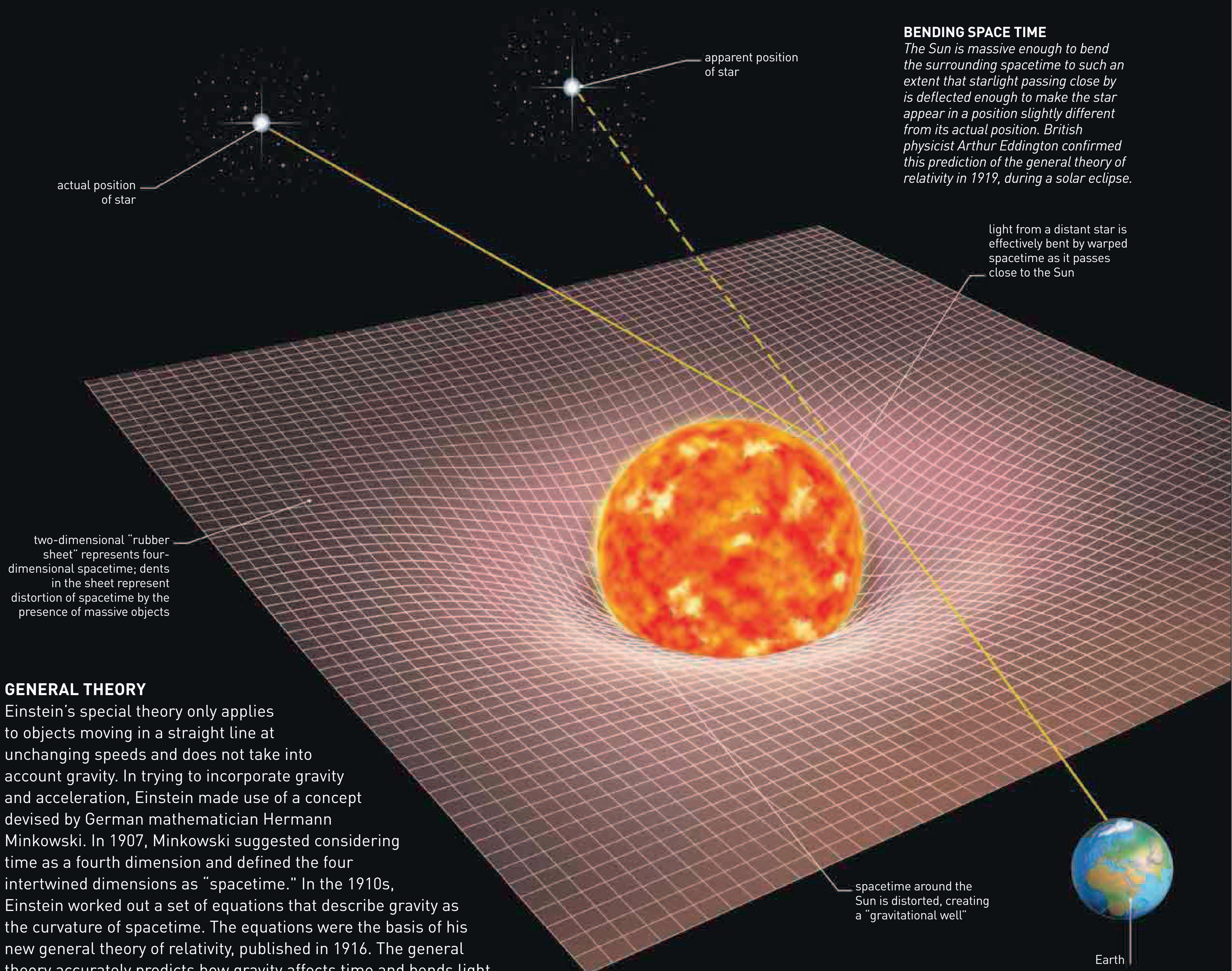
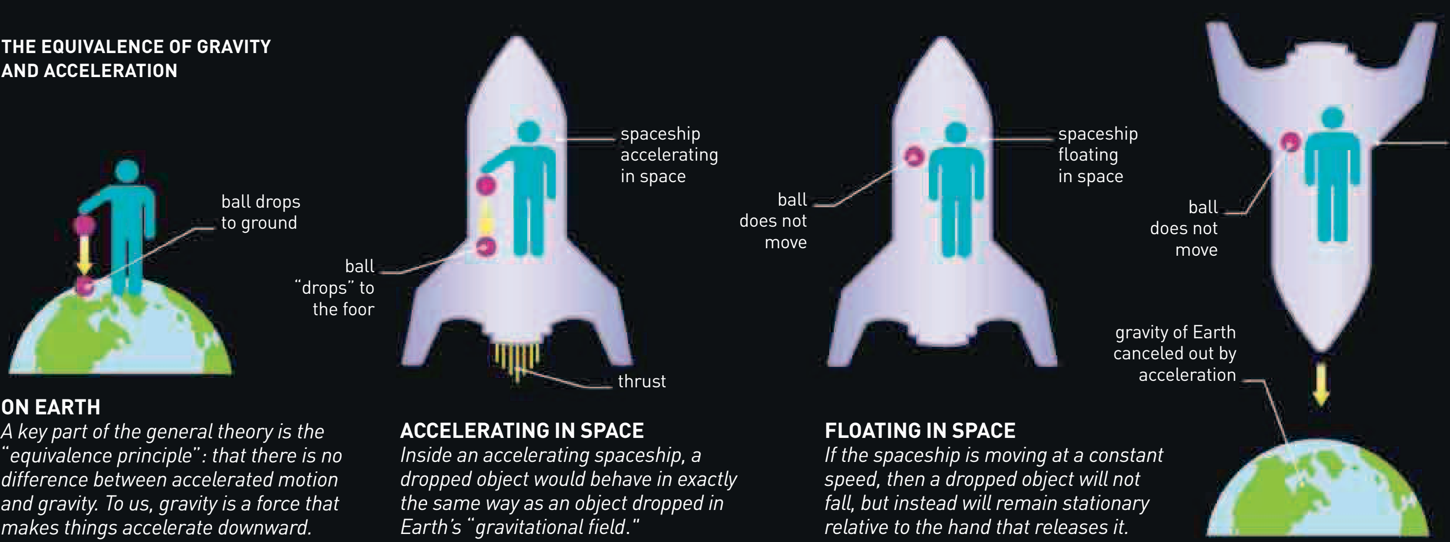



**BENDING SPACE TIME**
*The Sun is massive enough to bend the surrounding spacetime to such an extent that starlight passing close by is deflected enough to make the star appear in a position slightly different from its actual position. British physicist Arthur Eddington confirmed this prediction of the general theory of relativity in 1919, during a solar eclipse.*

## GENERAL THEORY

Einstein's special theory only applies to objects moving in a straight line at unchanging speeds and does not take into account gravity. In trying to incorporate gravity and acceleration, Einstein made use of a concept devised by German mathematician Hermann Minkowski. In 1907, Minkowski suggested considering time as a fourth dimension and defined the four intertwined dimensions as "spacetime." In the 1910s, Einstein worked out a set of equations that describe gravity as the curvature of spacetime. The equations were the basis of his new general theory of relativity, published in 1916. The general theory accurately predicts how gravity affects time and bends light.

**THE EQUIVALENCE OF GRAVITY AND ACCELERATION**




**ON EARTH**
*A key part of the general theory is the "equivalence principle": that there is no difference between accelerated motion and gravity. To us, gravity is a force that makes things accelerate downward.*

**ACCELERATING IN SPACE**
*Inside an accelerating spaceship, a dropped object would behave in exactly the same way as an object dropped in Earth's "gravitational field."*

**FLOATING IN SPACE**
*If the spaceship is moving at a constant speed, then a dropped object will not fall, but instead will remain stationary relative to the hand that releases it.*

**FALLING TO EARTH**
*Inside a spaceship in freefall, an object will accelerate downward at the same rate as the spaceship. There is no difference between accelerated motion and gravity.*